Practice workbook · Where a rational curve crosses the axes

Finding the coordinates where a curve y = (ax + b)/(cx + d) meets the x-axis and the y-axis.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-GRA-05

1. Question 3 · part c · 2 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

3

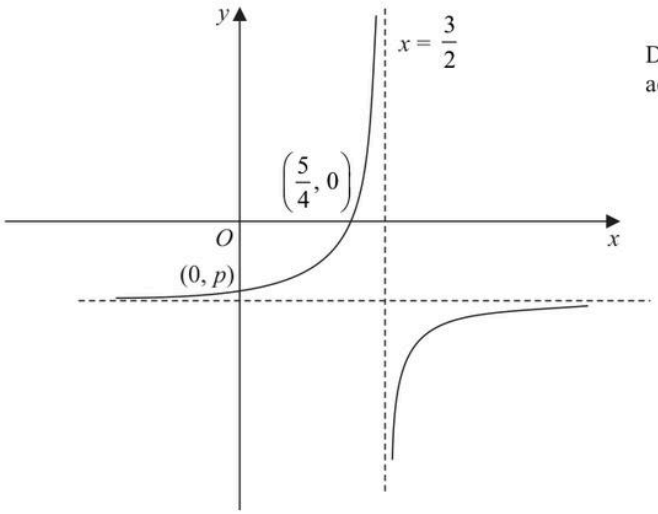


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of the curve with equation

$$y = \frac{a - bx}{cx - d} \qquad x \neq \frac{d}{c}$$

where a, c and d are prime numbers and b is an integer.

The asymptote to the curve that is parallel to the y-axis has equation $x = \frac{3}{2}$

(a) Write down the value of c and the value of d

(2)

The curve crosses the x-axis at the point $\left(\frac{5}{4}, 0\right)$

(b) Find the value of a and the value of b

(2)

The curve crosses the y-axis at the point $(0, p)$ where p is a rational number.

(c) Find the value of p

(2)

(d) Find an equation of the asymptote to the curve that is parallel to the x-axis.

(1)

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1. Question 3 · part c · 2 marks · Working and answer

Working and answer

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2. Question 3 · part c.i · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c.i only.

3 Curve C has equation $y = \dfrac{ax + 3}{1 - 2x}$ where $x \neq \dfrac{1}{2}$ and a is a constant.

The asymptote to C that is parallel to the x-axis has equation $y = 4$

(a) Find the value of a

(2)

(b) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

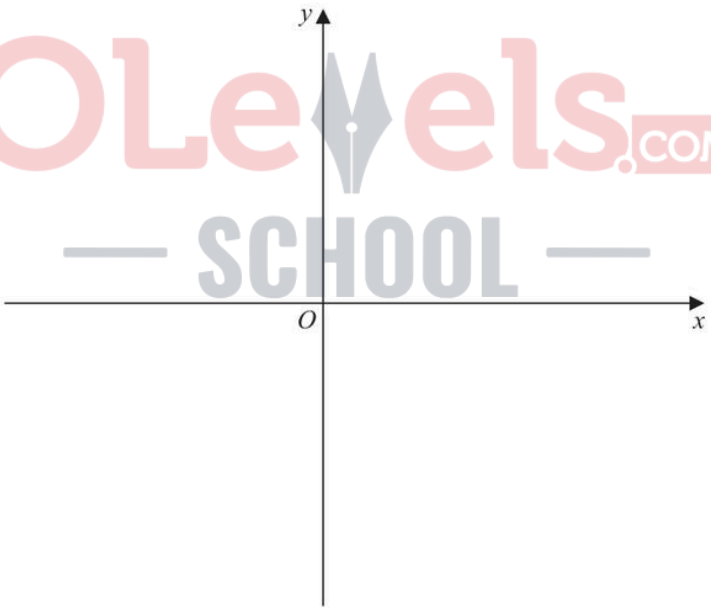
(c) Find the coordinates of the point where C crosses

(i) the x-axis, (ii) the y-axis.

(2)

(d) Using the axes below, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(4)



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P 7 1 8 1 7 A 0 8 3 2

2. Question 3 · part c.i · 2 marks · Working and answer

Working and answer

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3. Question 10 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

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3. Question 10 · part b · 2 marks · Working and answer

Working and answer

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4. Question 10 · part b · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

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4. Question 10 · part b · 2 marks · Working and answer

Working and answer

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5. Question 3 · part c.i · 0 marks

2021 JUN · Paper 01

Further practice · Attempt part c.i only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

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5. Question 3 · part c.i · 0 marks · Working and answer

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6. Question 3 · part c.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c.ii only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

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6. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

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7. Question 3 · part b.i · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b.i only.

3 Curve S has equation $y = 2 - \frac{5}{2x+1}$ where $x \neq -\frac{1}{2}$

(a) Write down an equation of the asymptote to S that is parallel to

(i) the y-axis (ii) the x-axis

(3)

(b) Find the coordinates of the point where S crosses

(i) the y-axis (ii) the x-axis

(2)

(c) Using the axes on the page opposite, sketch S, showing clearly the asymptotes and the coordinates of the points where S crosses the coordinate axes.

(3)

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7. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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8. Question 11 · part b.i · 0 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b.i only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

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8. Question 11 · part b.i · 0 marks · Working and answer

Working and answer

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9. Question 11 · part b.ii · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b.ii only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

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9. Question 11 · part b.ii · 2 marks · Working and answer

Working and answer

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10. Question 10 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

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10. Question 10 · part b · 2 marks · Working and answer

Working and answer

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11. Question 9 · part b.i · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b.i only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

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11. Question 9 · part b.i · 2 marks · Working and answer

Working and answer

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12. Question 10 · part a · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

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12. Question 10 · part a · 2 marks · Working and answer

Working and answer

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13. Question 10 · part a.i · 0 marks

2024 NOV · Paper 02

Further practice · Attempt part a.i only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

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13. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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14. Question 10 · part a.ii · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a.ii only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

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14. Question 10 · part a.ii · 2 marks · Working and answer

Working and answer

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15. Question 10 · part c · 1 mark

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

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15. Question 10 · part c · 1 mark · Working and answer

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16. Question 4 · part c · 2 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

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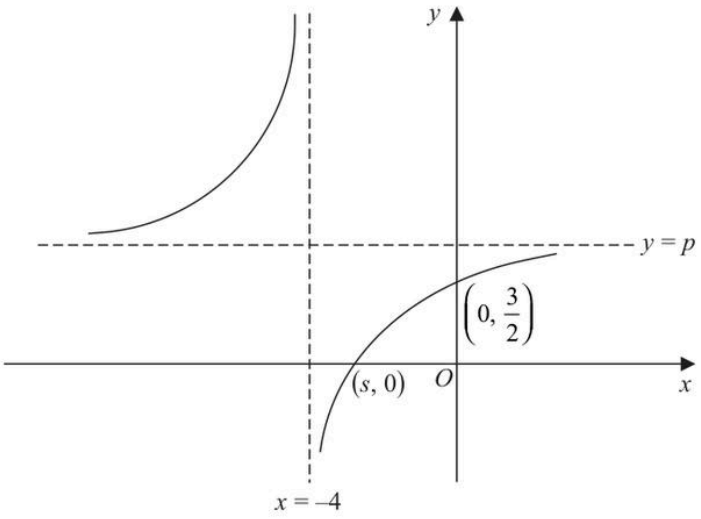


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation

$$y = \frac{2x + q}{x + r} \qquad x \neq -r$$

where q and r are integers.

The asymptote to C that is parallel to the y-axis has equation $x = -4$
The asymptote to C that is parallel to the x-axis has equation $y = p$

(a) Write down

(i) the value of p

(ii) the value of r

(2)

Given that C crosses the y-axis at the point with coordinates $\left(0, \frac{3}{2}\right)$

(b) find the value of q

(2)

Given that C crosses the x-axis at the point with coordinates $(s, 0)$

(c) find the value of s

(2)

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16. Question 4 · part c · 2 marks

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Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 6 marks)

16. Question 4 · part c · 2 marks · Working and answer

Working and answer

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